Varsha Rajawat

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EDUCATION:

Rajasthan Technical University **Aug 2015 – Jun 2019**
Government Mahila Engineering College, Ajmer
Bachelor of Technology in Information Technology

Relevant Courses: Engineering Mathematics (Linear Algebra, Multivariable Calculus, Vectors, Complex numbers)
Fundamental of Computer programming, Object Oriented Programming, Advanced Data Structures,
Advance Data Base Management Systems, Data Mining & Ware Housing

EXPERIENCE:

Data Scientist at Tiger Analytics, Bangalore **Feb 2022 – Present**

Trade Promotion & Optimization:

The project aimed to optimize the pricing and promotions of the products across the Poland Pet market for 6 categories to reduce the risk incurred by the RCPG client due to arbitrage and further recommend the selling price and the promotions with optimized slots to the retailers.

- Designed and implemented a linear optimization Algorithm, pulp against the constraints/objective functions dealing with KPIs like Net Sales Value, Gross Sales Value, Gross Margin to get the best prices and promotions.
- Three types of price elasticity values – own elasticity, cross internal elasticity, cross external elasticity, and three types of temporary price reductions – own TPR, cross internal TPR, and cross external TPR, wereused to define constraints that accounted for the change in the demand with change in the prices.
- Developed Bayesian Regression models at a retailer-category-product level to identify the elasticity values.
- Further tested multiple logics to improve the identification of the competitor products (for cross elasticity values and cross TPR values) and used features like seasonality, promotion variables, Holiday data, etc. to improve the model compliance – higher r2_score and lower MAPE.
- Designed and built descriptive pricing and promo PowerBI dashboard to visualize sales and promotion performances among various retailers, manufacturers, categories, etc.

Senior Analyst at Tiger Analytics, Bangalore

Market Media Mix:

Developed a Market Media Mix Model for a UK food market to evaluate the true incrementality of marketing investments, optimize Return on Investment (ROI), and balance spending across traditional, digital, national media, and shopper marketing channels.

- Processed, cleaned, and analyzed extensive sales and media data from major retailers such as Amazon, Target, and Walmart to uncover insights on media channel performance.
- Built and trained predictive models using Lasso Regression and Bayesian Techniques, with R-squared as the primary performance metric, using features like ad-stock, clicks, searches, etc. These models quantified the impact of media spending on sales.
- Analyzed the synergy effect of various marketing channels and ran over 20,000 models on different combinations to evaluate their performance individually and combined.
- Generated and interpreted response curves to identify the optimal spending range, ensuring maximum ROI and efficient allocation of resources.

On-Demand Delivery Assortment Dashboard:

Developed a comprehensive Portfolio Dashboard for a UAE food market to provide actionable insights into product financial and market performance, aiding in strategic decision-making for portfolio management, assortment planning, and promotional strategy development.

- Worked with Azure Databricks and used PySpark to perform analytics.
- Processed, cleaned, and analyzed extensive client and competitor sales and financial datasets to evaluate product distribution, sales performance, pricing trends, and portfolio dynamics.
- Designed and implemented a Five-Star Framework using five distinct matrices to identify high-performing and low-performing SKUs based on matrices NSV, RSV, GSV, etc. facilitating informed prioritization.
- Developed a Quadrant Analysis Framework to categorize SKUs into four strategic categories—Prioritize, Optimize, Invest to Grow, and Rationalize—based on sales and distribution performance.
- Developed CI/CD pipelines using Azure DevOps to build and deploy code into the production environment.

Associate at Cognizant Technology Solutions, Bangalore **Jan 2022– Feb 2022**

Customer Enquiry Chatbot:

Developed an intelligent chatbot for a leading USA-based insurance firm to streamline business-customer interactions by automating responses, delivering timely notifications, and conducting surveys.

- Data Preprocessing: Implemented a comprehensive pipeline to process, clean, tokenize, and vectorize customer query data using the RASA framework.
- Intelligent Rule Design: Designed and implemented custom labels and rules to ensure accurate and context-aware responses to customer inquiries.
- Entity Extraction: Utilized regular expressions and natural language processing techniques to identify and extract key entities from customer queries, enhancing the chatbot's understanding of user intent.

Programmer Analyst at Cognizant Technology Solutions, Bangalore **Dec 2020 – Jan 2022**

Denial Claim Prediction:

Developed a predictive model for an orthopedic product provider to address insurance claim denials caused by missing or incorrect information. The solution enabled proactive identification of potentially denied claims, improving operational efficiency, and reducing claim rejections.

- Data Analysis: Conducted extensive Exploratory Data Analysis (EDA) to uncover patterns, trends, and key insights within the data.
- Data Balancing: Addressed class imbalance in the dataset using advanced data reduction techniques, ensuring robust model performance.
- Feature Engineering: Applied statistical methods, including Weight of Evidence (WOE), Information Value (IV), and correlations, and transform textual data into meaningful numerical features.
- Model Development: Built and trained a Random Forest classification model and ran multiple models using GridSearchCV for best parameters to predict denial probabilities accurately.
- Model Evaluation: Assessed model performance using metrics such as F1 score, gain charts, and lift charts to ensure reliability and scalability.

Programmer Analyst Trainee at Cognizant Technology Solutions, Bangalore **Dec 2019 – Dec 2020**

Account Receivables Inventory Prioritization:

Designed and implemented a tool for an orthopedic product provider to optimize claim prioritization and allocation in the Accounts Receivables process, enhancing efficiency and streamlining team workflows.

- Requirements Analysis: Gathered and analyzed detailed business rules to identify key criteria for claim prioritization, aligning with organizational objectives.
- Tool Development: Developed a custom tool to automate the prioritization of claims based on predefined rules and business requirements.
- Optimized Allocation: Balanced workloads by grouping and distributing claims in equal ratios across team members, ensuring efficient resource utilization and workflow management.

RELEVANT PROJECTS AND TRAINING:

Python (Core + Advance) + Data Science Training | GRRAS Solutions Pvt. ltd., Jaipur **May 2018 – Jul 2018**

GRRAS Solutions specializes in Python and Data Science among other courses for In-house training, Industrial/Internship training, Online Learning, and Corporate Training.

Housing Price Prediction – Regression [link]

Developed a predictive model for housing prices using features such as location, property size, and number of rooms. Implemented advanced machine learning algorithms, including Random Forest, Gradient Boosting (XGBoost, LightGBM), and Neural Networks, to improve accuracy. Conducted data cleaning, feature engineering, and hyperparameter tuning. Evaluated model performance using metrics like R-Square, MAE, and RMSE. Key insights included the significance of location and property size in price determination, providing actionable insights into market trends.

ACADEMIC AND INDEPENDENT PROJECTS

Spam and Non-Spam Email Prediction - Classification [link]

Developed a machine learning model to classify emails as spam or not spam using historical email data. Utilized CountVectorizer for text vectorization and implemented a Naive Bayes classifier to achieve robust classification. Performed data preprocessing, feature extraction, and predictions.

TECHNICAL SKILLS

- **Machine Learning:** ScikitLearn, Scipy, Statsmodels, MLlib, XGBoost, Keras, Tensorflow, NLTK, NumPy, Pandas
- **Programming:** Python, R, C++, PySpark, SQL
- **Software:** Power BI, Excel, Databricks, Microsoft Azure, SQL Server, Visual Studio

ACTIVITIES AND AWARDS

- **Media Committee Secretary** | Government Mahila Engineering College, Ajmer
  Organized and ensured smooth conduction of several photography and videography events.
- **Mime, Epistemico |** Cultural Fest, Government Mahila Engineering College, Ajmer
  Directed, wrote, and managed a team mime act, leading to a winning performance at college fest competition.
- **Sports, Taksh |** Sports Fest, Government Mahila Engineering College, Ajmer
  Captioned, managed multiple sports team such as volleyball team, throwball team, table tennis team etc.